

FFT Mini SoC

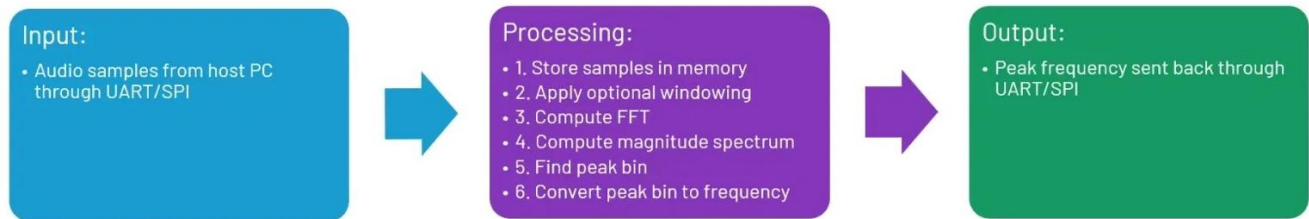
Architecture & HW/SW Partitioning

Discussion

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Introduction

The SoC receives a block of audio samples through a UART or SPI, computes the FFT spectrum, identifies the dominant frequency component, and sends the peak frequency result back to the host.



Hardware Software Partitioning

1. Storing samples in memory: the CPU could read the data from the UART receiver and write it to memory (SW)
2. Applying optional windowing: it requires an element wise vector multiplication so its better to perform it on hardware with the CPU providing the controls for optional windowing (HW)
3. Computing FFT: requires repetitive multiplications and additions so its better on hardware (HW)
4. Computing magnitude spectrum: requires calculating the square root and squaring elements for each bin so its better on hardware (HW)
5. Find peak bin: simply comparing the frequency bins and finding the max, could be handled by software (SW)
6. convert peak bin to frequency: also a simple operation of mapping the peak bin into the actual frequency a simple operation for software (SW)

SoC Architecture

Based on the hardware software partitioning the SoC architecture should look something like this:

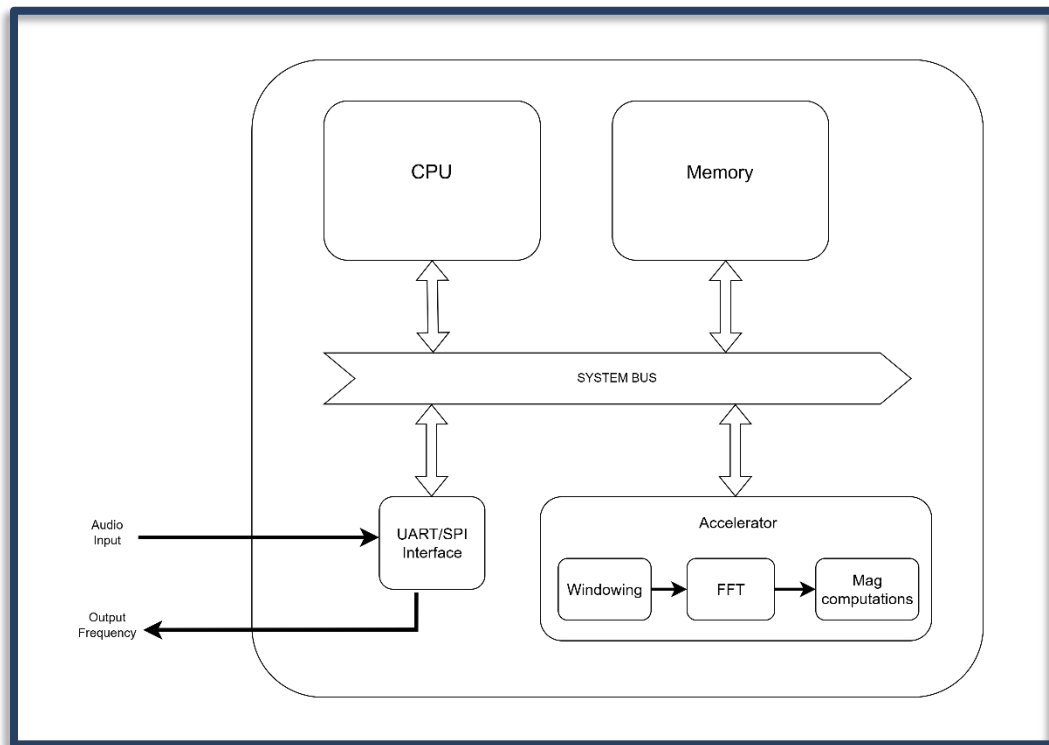


Figure 1 : SoC Architecture

System Application Pseudo Code

- 1-receive from uart/spi
- 2-store data in mem
- 3-send data to accelerator + optional windo control signal
- 4-wait for the mag computations results of accelerator
- 5-find peak bin
- 6-convert peak bin into frequency
- 7-send output frequency to uart/spi
- 8-repeat for next audio sample block